

Company: Samsung R&D Institute India-Bangalore (SRI-B)

Internship: 2 months (Virtual) – last week on campus (Bengaluru)

Online Test:

contained 3 questions

1) Bitwise operator question

2) sum up alternative nodes -

<https://www.geeksforgeeks.org/print-alternate-nodes-from-all-levels-of-a-binary-tree/>

3) array based question – it was easy but optimal solution was expected (Time – $O(N)$ and space – $O(1)$)

Interview (Virtual): around 1 hr

1) No questions asked regarding projects, was asked about introduction and then the interview directly asked what's my favorite data structure and started asking questions .

2) questions asked – LinkedList : reverse a linkedlist , inserting a node in a sorted linkedlist , deletion of a node , finding the intersection point of two linkedlist (

<https://leetcode.com/problems/intersection-of-two-linked-lists/description/>)

Searching a node in binary tree

3) OS concepts – Threading, Multi – threading , Memory Management techniques

4)OOPS concept – 4 pillars (abstraction, encapsulation ,polymorphism , inheritance)

5)Testing your ability to think and approach a problem –
Interviewer gave a sample use case and asked me to give a solution.

The question was something like:

There is an array which has memory spaces allocated to it, when a program utilizes the CPU the spaces in the array are allocated to each program from the array. Write an algo to optimize the memory allocation efficiently.

Sol: Use a Boolean variable ,Mark the used index(Utilized spaces) of array as false and keep track of free spaces and check for adjacent index which is also free and expand the available free places to make it as a big chunk of available memory to avoid segmentation .

Last year they asked Dp – egg drop question and Graph as well so it's better to be prepared with all the DSA topics (especially LinkedList , Tree, Graph and DP) it all depends on the interview !

GOODLUCK !